

A Supervisory Portfolio Governance Framework: Composite Instability Detection, Deterministic Regime Switching, and Conversational Explainability

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ARTICLE INFO

Keywords:

portfolio governance
instability detection
regime switching
shrinkage estimation
explainable AI
drawdown protection

ABSTRACT

Conventional mean-variance optimizers cannot simultaneously deliver robust out-of-sample allocation under estimation uncertainty and the fully auditable decision records required by MiFID II, GDPR Article 22, and Basel III. This paper proposes a supervisory governance framework that addresses both requirements through three formally defined components: a **Composite Instability Index** I_t aggregating cross-sectional volatility, mean pairwise correlation, and covariance matrix drift measured by the Frobenius norm; a deterministic **Regime Operator** $\mathcal{R}(I_t)$ that maps I_t to a binary, fully reproducible portfolio construction rule; and a **Governance Stability** metric quantifying mean ℓ_1 weight change across rebalancing periods. The covariance drift component is novel in portfolio governance and exhibits a 6.02-fold mean-level separation between market states. Evaluated across 51 non-overlapping quarterly rolling windows on 19 US equities (2012–2024), the framework reduces mean maximum drawdown by 38.1% versus a static mean-variance baseline ($t = 5.67$, $p < 0.001$), robust to COVID-19 exclusion ($p < 0.001$, $n = 47$), invariant to risk-aversion parameter choice over a tenfold range, and stable within 2.0 percentage points across the full threshold grid. Governance Stability improves by 53% and effective diversification increases 4.3-fold. Cross-universe validation across five asset universes confirms positive and statistically significant drawdown improvement in every case. A post-hoc explanation interface allows portfolio supervisors and compliance officers to query any governance decision in plain language, without access to or influence over the underlying computation.

1. Introduction

The position of portfolio optimization in quantitative finance is a little odd: the theory, based on Markowitz's quadratic program (Markowitz, 1952), is well developed, while the out-of-sample performance is still disappointing after many decades. What is the problem? Optimality can only be proved in the presence of exact expected returns and covariances, whereas in reality, we can only estimate them from a finite set of noisy data, and the optimizer becomes an error maximizer.

The evidence is consistent and cross-validated. Chopra and Ziemba (1993) has shown that errors in expected return are eleven times more detrimental to portfolio quality than similar errors in covariance matrices. Best and Grauer (1991) has demonstrated significant effects of small return input perturbations on optimal weights. The main problem with empirical studies is raised in DeMiguel et al. (2009), where none of 14 different optimisation methods and 7 datasets managed to outperform the naive equal weight strategy in terms of out-of-sample Sharpe ratio.

Structurally independent pressure has developed in parallel with the estimation problem. Financial regulation now directly addresses the systems of automated allocation. In the above regulations, MiFID II (European Parliament, 2014), the General Data Protection Regulation (GDPR) (European Parliament, 2016), and Basel III (Basel Committee on Banking Supervision, 2011), collectively, make mandatory the documentation, reproducibility, and explainability of every investment decision taken through the system on demand. This is not aspirational but rather binding. What is demanded is not a performance metric but rather the ability to identify the signal observed, the rule applied, and

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Replication code and data retrieval scripts are publicly available at https://github.com/jithendra259/fintech_chatbotk.

the results that would have been obtained from alternative input sets. This is not something that can be added as an afterthought but rather something that has to be designed-in from the start.

These forces have been the focus of the literature, although in an isolated manner. The shrinkage-based methods such as the Ledoit–Wolf (Ledoit and Wolf, 2004b,a) and the James–Stein estimation (James and Stein, 1961; Jorion, 1986) significantly alleviate the noise in the inputs but do not address the behavior of the allocation system when the estimation context deteriorates. The Markov regime-switching models (Hamilton, 1989; Ang and Bekaert, 2002, 2004) improve the allocation system in the face of changing markets (Guidolin and Timmermann, 2007; Costa and Kwon, 2019; Nystrup et al., 2019), but they produce continuous regime probabilities that require a human judgment interface before the actual portfolio weight calculations are performed, thus reinstating the very type of undocumented judgment the regulation aims to address. Explainable AI models (Arrieta et al., 2020; Doshi-Velez and Kim, 2017; Adadi and Berrada, 2018) provide a wealth of post-hoc explanations; yet, they are not part of a decision-making framework that satisfies the auditability requirements of financial regulation.

Research Gap

Although significant progress has been made in the three streams, no model has yet been developed that combines deterministic portfolio governance, auditability, and on-demand explainability within a formally defined expert system and has been validated out-of-sample over a long and multi-stressful time period.

In this paper, we propose a supervisory expert system in portfolio governance where auditability is considered a main requirement in portfolio governance. The three parts of the model correspond to the three regulatory functions.

Detection: The Composite Instability Index I_t aggregates three Z-score-based indicators: cross-sectional volatility, mean pairwise correlation, and covariance matrix drift as quantified through the Frobenius norm of successive estimation windows' difference. Following Billio et al. (2012) and Acharya et al. (2017), who showed through their studies that structural co-movement changes are early signs of systemic stress, the drift component δ_t is, on average, 6.02 times higher in windows of high instability compared with windows of low instability, the strongest of the three variables and the first formalization of this variable for portfolio governance.

Decision: The Regime Operator maps the Composite Instability Index to one of two possible portfolio construction instructions with a single scalar comparison with no degrees of freedom left to third parties. Given the same input and threshold, any third party will necessarily derive the same instruction as the system, thus satisfying the audit requirements of MiFID II (European Parliament, 2014), GDPR Article 22 (European Parliament, 2016), and Basel III (Basel Committee on Banking Supervision, 2011).

Explanation: A large language model is granted read-only access to the committed governance record and is used to translate the decision into narrative form on demand. The separation of the model from the computation is structural rather than prompt-based and is therefore verifiable.

This system architecture is a knowledge-based expert system with deterministic rules activated by market signals to produce fully auditable allocation decisions. This is directly within the scope of the core subject area of expert and intelligent systems research (Arrieta et al., 2020; Wooldridge, 2009).

Evaluation of the framework across 51 non-overlapping quarterly time windows on 19 U.S. equities from January 2012 to October 2024 shows that the framework decreases the mean maximum drawdown by 38.1% compared to a static mean variance approach (Markowitz, 1952) in the primary asset universe ($t = 5.67$, $p < 0.001$), while the cross-universe mean is approximately 32.5%. This holds even when excluding the COVID-19 period ($p < 0.001$, $n = 47$), is insensitive to risk aversion across a tenfold increase in risk aversion levels, and stable across the full grid of thresholds within 2.0 percentage points. Governance Stability is increased by 53%, effective diversification is increased 4.3-fold, and the cross-universe drawdown improvement is from +12.4% to +70.0%.

This paper contributes five specific contributions: Frobenius norm covariance drift as a formally specified governance signal, achieving a mean level separation of 6.02 between states; Regime Operator, a deterministic switching mechanism directly compliant with MiFID II (European Parliament, 2014), GDPR Article 22 (European Parliament, 2016), and Basel III (Basel Committee on Banking Supervision, 2011), eliminating the discretionary translation layer of probabilistic regime models (Hamilton, 1989; Ang and Bekaert, 2002, 2004); Governance Stability as a new evaluation dimension; empirical results on Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004b,a) under conditions of extreme instability; and a conversational governance interface, where the isolation of the language model's computation is verifiable based on the system architecture, in addition to the verification of the instructions in the prompt.

2. Literature Review

2.1. Mean-Variance Optimisation: A Framework Under Strain

Modern portfolio theory starts from a deceptively simple idea. Markowitz (1952) showed that rational investors should care not only about return, but also about the trade-off between return and variance. That insight gave portfolio choice a crisp mathematical form and, in doing so, helped define a large part of modern quantitative finance. Yet the practical weakness of the framework has been evident for a long time: when the inputs are estimated with error, out-of-sample performance is often disappointing.

The difficulty is straightforward to describe. Both the expected return vector and the covariance matrix must be estimated from historical data, and those estimates are inevitably noisy. The scale of that noise depends on the length of the estimation window and the number of assets being modelled. In a classic critique, Michaud (1989) argued that the mean-variance optimiser does not simply inherit input error; it can amplify it. Assets whose observed returns are flattered by sampling variation may be overweighted, whereas assets with weak realised histories may be pushed aside even when those histories are only partially informative about future performance.

The consequences also depend on where the largest estimation errors occur. Chopra and Ziemba (1993) showed that errors in expected returns are far more damaging to portfolio utility than comparable errors in the covariance matrix. This asymmetry matters in practice because many stabilisation methods focus mainly on covariance regularisation and therefore address only part of the problem. From a related angle, Best and Grauer (1991) showed that even small changes in return inputs can produce large jumps in optimal portfolio weights. Such sensitivity makes the optimiser's output harder to interpret and harder still to defend in settings where clients, supervisors, or compliance staff expect a clear and justifiable rationale.

Attempts to address this problem have produced mixed evidence. One of the most influential empirical comparisons in the literature is DeMiguel et al. (2009), who evaluated fourteen portfolio strategies across seven real-world datasets using rolling out-of-sample tests. In that study, no optimised strategy consistently outperformed the simple equal-weight benchmark in Sharpe ratio terms. That result has been debated at length, but it remains central to the question motivating this paper: under what explicitly specified conditions does an analytical allocation rule improve on a naive benchmark, and how should the system behave when those conditions are not met?

2.2. Shrinkage Estimation: Progress and Residual Limitations

The prevailing technical response to estimation noise in portfolio construction has been shrinkage. Statistically, this approach traces back to James and Stein (1961), who showed that the sample maximum-likelihood estimator is inadmissible under squared-error loss when three or more parameters are estimated jointly. In practical terms, better estimates can often be obtained by pulling noisy observations toward a common target. Although the result seemed counterintuitive when first introduced, it has since been validated across a wide range of settings and loss functions.

One of the earliest direct applications of this idea to portfolio management was Jorion (1986), who used the Bayes-Stein estimator to shrink individual expected returns toward the cross-sectional average. The intuition is straightforward: unusually high historical returns are treated more cautiously, which helps limit the error-amplification problem identified by ?.

On the covariance side, the main practical breakthrough came from Ledoit and Wolf (2004b). Earlier methods often relied on cross-validation to select the shrinkage intensity, an approach that was computationally expensive and could itself introduce additional estimation noise. Ledoit and Wolf (2004b) derived the optimal shrinkage intensity in closed form, and Ledoit and Wolf (2004a) recast the same logic for a practitioner audience. Together, these papers established the Ledoit-Wolf estimator as a standard tool for covariance regularisation in asset management. Ledoit and Wolf (2017) later extended the framework to nonlinear shrinkage of individual eigenvalues, which performs especially well in the high-dimensional settings common in modern institutional portfolios.

The statistical properties of shrinkage estimators are well understood. What the literature has not addressed is a different and arguably more important governance question: what should an allocation system do when the covariance structure is changing faster than any estimator — regularised or otherwise — can reliably track? A shrinkage prior calibrated on a stable, low-correlation window is a poor starting point when correlations have since spiked and the covariance matrix has rotated substantially. The empirical evidence in this paper is specific: covariance matrix drift, measured by the Frobenius norm of successive window differences, is 6.02 times higher on average during high-instability periods than during stable ones. At that rate of structural change, the question is not how tightly to regularise — it is whether mean-variance optimisation remains an appropriate allocation mechanism at all.

2.3. Regime-Switching and the Auditability Gap

A distinct body of research has approached the same underlying problem — non-stationarity in financial markets — through explicit modelling of discrete state changes. Hamilton (1989) demonstrated that many economic time series exhibit abrupt distributional shifts that smooth-evolution models cannot accommodate, and that a hidden Markov framework captures these shifts substantially more accurately. The asset allocation implications were developed by Ang and Bekaert (2002, 2004), who showed that ignoring regime structure leads to economically significant utility losses, particularly because correlations tend to rise sharply during downturns (Longin and Solnik, 2001) — precisely when the diversification benefits promised by a static allocation are least likely to materialise. Guidolin and Timmermann (2007) showed that left-tail risk can be reduced by portfolio rebalancing conditional on estimated regime states. Similar support appears in later work by Costa and Kwon (2019), Nystrup et al. (2019), and Wang et al. (2020), who examine different asset classes and sample periods.

Probabilistic regime models remain attractive because they are statistically disciplined, responsive to changing market conditions, and naturally suited to state-dependent allocation. Even when estimation improves, however, their main governance weakness does not disappear. A Markov regime model typically produces a posterior probability, for example 0.73, of being in a high-volatility state. That output is informative, but it is not yet a portfolio instruction. An additional rule is needed to convert the posterior into actual weights, whether by blending two target portfolios according to the posterior or by switching only when the probability exceeds a threshold. Both choices involve judgment, and neither is uniquely implied by the probabilistic output. As a result, two risk managers looking at the same posterior could still arrive at materially different allocations, often without a translation rule that a regulator can fully reconstruct.

In a regulatory context, this distinction matters. European Parliament (2014) Article 25 and European Parliament (2016) Article 22 do not simply favour decisions that appear reasonable; they require decisions that can be reproduced from documented inputs and rules. Deterministic threshold comparison addresses that need directly. A single scalar input is converted into a binary portfolio instruction, and once the rule is specified there is no residual discretion. Any third party with access to the same data should therefore reach the same allocation decision. That is the kind of audit trail the present framework is designed to provide.

2.4. Composite Instability Signals: Independent Validation

The composite instability index presented here combines three signals that are supported by prior empirical work.

Cross-sectional volatility. Engle (1982) and Bollerslev (1986) showed that equity volatility exhibits time-dependent patterns, such that high volatility today can help predict volatility tomorrow over horizons relevant for portfolio rebalancing. In governance terms, average cross-sectional volatility is useful because it acts as a *forward-looking* signal: it contains information about changes in the return distribution rather than merely describing what happened in the most recent period.

Mean pairwise correlation. Longin and Solnik (2001) found that equity correlations tend to rise during bear markets, a result that has been confirmed repeatedly in later studies. The governance relevance is direct. A mean-variance portfolio constructed using a shrinkage covariance estimated in a low-correlation regime embeds diversification assumptions that may fail exactly when they are needed most. In the dataset used here, the share of equity pairs with 60-day rolling correlations above 0.70 increased by 62.5% during the stress period relative to the training period.

Covariance matrix drift. Billio et al. (2012) and Acharya et al. (2017) show that structural changes in co-movement across the full correlation matrix are important for detecting periods of rising market stress. Their findings suggest that univariate volatility measures and aggregate correlation metrics cannot fully capture these shifts on their own. In this paper, covariance matrix drift is operationalised as the Frobenius norm of successive covariance-window differences, producing a daily scalar that detects rapid full-matrix rotation events. To our knowledge, this measure has not previously been formalised as a governance indicator in portfolio management. The result that it is 6.02 times larger in high-instability regimes than in stable regimes suggests that it adds governance information beyond the two more established signals.

2.5. Expert Systems and Explainable AI in Financial Decisions

The domain in which this paper is situated is the design of expert systems for financial decision support — a tradition rooted in rule-based inference and now increasingly concerned with making machine reasoning legible to the humans who must act on, or be held accountable for, its outputs.

The earliest expert systems in finance were built around explicit if-then rules whose logic was transparent by construction (?). In the language of contemporary XAI research, that property is described as ante-hoc interpretability. It is especially valuable in regulatory settings because the decision logic is specified before any decision is made, rather than reconstructed after the fact from a model that was never designed to be explained. The Regime Operator in this paper follows that tradition through a formally defined threshold rule that produces all outcomes directly from its specification, without requiring additional methods to trace how a decision was reached.

Current XAI research attempts to cover the full interpretability landscape. Arrieta et al. (2020) provide one of the first comprehensive taxonomies and distinguish between two broad forms of interpretability. Their framework also clarifies the settings in which each approach is most appropriate. In the present context, the key point is that ante-hoc interpretability is especially important in high-stakes financial decisions whenever it can be achieved without reducing decision quality. At the same time, much of the academic XAI literature emphasises explanation formats such as SHAP values, feature importance scores, and counterfactual examples. As Doshi-Velez and Kim (2017) note, these tools do not fully solve the practical problem of explanation when the audience consists of compliance teams or regulators who may require additional technical training to use them effectively. The critique by Adadi and Berrada (2018) reinforces this point: in the absence of accepted standards for what counts as an adequate explanation, genuine uncertainty remains for both system designers and the regulators responsible for evaluating them.

Large language models have recently emerged as a potential bridge between technical decision logic and human-readable audit records. Wu et al. (2023) demonstrated that LLMs grounded in structured quantitative inputs produce substantially more accurate financial narrative than ungrounded generation, with significantly lower rates of factual confabulation. Li et al. (2023) identified post-hoc narration of automated allocation decisions as a high-value and largely unexplored application, distinct from the prediction and classification tasks that have dominated LLM research in finance. The FinAgent architecture of Zhou et al. (2024) demonstrated that language model agents can be integrated into active trading pipelines, but in that design the model participates directly in the decision process. The architecture developed here takes a different position: the language model is permitted only to explain governance records that have already been committed and cannot be modified. This isolation is structural — enforced by the pipeline’s execution sequence rather than by prompt-level instruction alone — ensuring that the model’s narration capability has no pathway back to upstream portfolio computation.

2.6. Research Gap and the Positioning of This Work

The five streams outlined above have each advanced the portfolio governance problem, but they have not yet been brought together in a single integrated solution.

Shrinkage estimation helps stabilise inference when the inputs are noisy, but it does not provide a mechanism for detecting when the underlying estimation conditions themselves begin to deteriorate, nor does it offer guidance on how the system should respond when that happens. Probabilistic regime-switching models provide useful state information, but their output is a continuous posterior that does not satisfy auditability requirements under European Parliament (2014) and European Parliament (2016) Article 22 until an additional translation step is introduced. The three instability signals discussed above — volatility clustering, correlation spiking, and covariance drift — have each been validated individually, yet the literature has not produced a formal composite index that incorporates a stability dimension and can function as an autonomous governance measure. Explainable AI has generated a rich set of interpretive tools, but those tools have not been embedded in a live governance framework that defines decision boundaries, isolates explanation from upstream computation, and evaluates performance under real stress episodes rather than only under ordinary market conditions. Expert systems for financial decision support have long relied on ante-hoc interpretability, but the literature still lacks a unified framework that combines modern covariance regularisation, composite regime detection, and natural-language explainability.

This paper addresses these five gaps jointly. The proposed system uses Ledoit-Wolf shrinkage (Ledoit and Wolf, 2004b) as its core allocation engine and combines it with a deterministic switching rule based on a formally defined composite instability index. It also introduces Governance Stability as an independent assessment category with its own formal definition and measurement procedure. A dedicated large language model, kept separate from the allocation engine, provides immediate plain-language explanations for governance decisions. The full framework is then evaluated as an expert decision-support system across 51 rolling quarterly windows and five asset universes from January 2012 to October 2024, covering both ordinary market conditions and major stress episodes such as the COVID-19 crash, the 2018 volatility shock, and the 2022 rate-driven sell-off.

3. Methodology

3.1. Data and Descriptive Statistics

The empirical dataset consists of daily adjusted closing prices for 19 US equities drawn from six sectors: Technology (AAPL, MSFT, GOOGL, AMZN), Healthcare (JNJ, PFE, UNH), Financial Services (JPM, BAC, GS, MS), Consumer Staples (PG, KO, PEP), Energy (XOM, CVX, COP), and Industrials (BA, CAT). The data are collected from Yahoo Finance with split and dividend adjustments included (?). The dataset contains 3772 daily returns for each asset over the period January 2010 to December 2024. The sector selection spans both defensive and cyclical businesses, allowing the governance framework to be evaluated under a wider range of volatility and correlation structures than would be possible in a single-sector sample.

The data are divided into two time intervals. The training period runs from January 2010 to December 2019 and contains 2515 trading days. This sub-period serves two purposes: (1) to provide the historical sample used to calibrate the *Z*-score normalisation procedure and (2) to supply the data needed to compute the Ledoit–Wolf shrinkage intensity (Ledoit and Wolf, 2004b,a) and the James–Stein shrinkage factor (James and Stein, 1961; Jorion, 1986). The rolling governance evaluation begins in January 2012, since the first governance cycle cannot be formed until a two-year warm-up period has been completed.

Table 1: Descriptive statistics for the 19 US equities in the sample (January 2010 to December 2024). Annualised return and annualised volatility are computed by scaling daily means and standard deviations by $\sqrt{252}$. Excess kurtosis is reported relative to the normal distribution benchmark of 3. All 19 assets reject the Jarque–Bera test of normality at the 0.1% significance level ($p < 0.001$).

Ticker	Sector	Ann. Ret. (%)	Ann. Vol. (%)	Skewness	Ex. Kurt.	JB <i>p</i>
AMZN	Technology	26.30	35.12	+0.42	8.21	<0.001
AAPL	Technology	24.06	30.45	−0.18	7.84	<0.001
UNH	Healthcare	23.88	28.33	−0.31	6.92	<0.001
BA	Industrials	20.05	34.87	−1.12	19.46	<0.001
MSFT	Technology	18.71	27.61	−0.22	7.13	<0.001
GOOGL	Technology	14.54	28.92	−0.19	6.78	<0.001
JPM	Financial Svcs	14.23	29.14	−0.44	11.23	<0.001
CAT	Industrials	12.02	29.76	−0.28	4.09	<0.001
JNJ	Healthcare	11.17	19.84	−0.33	7.46	<0.001
PEP	Consumer Staples	11.00	17.62	−0.41	8.33	<0.001
PFE	Healthcare	10.98	22.47	−0.19	4.62	<0.001
PG	Consumer Staples	10.32	17.43	−0.38	9.14	<0.001
KO	Consumer Staples	9.73	17.21	−0.27	9.87	<0.001
BAC	Financial Svcs	9.17	32.88	−0.51	11.74	<0.001
COP	Energy	8.23	34.21	−0.28	17.21	<0.001
CVX	Energy	7.96	28.94	−0.34	26.86	<0.001
MS	Financial Svcs	6.62	33.47	−0.42	12.38	<0.001
GS	Financial Svcs	4.24	30.12	−0.39	10.92	<0.001
XOM	Energy	3.33	27.43	−0.28	13.47	<0.001
<i>Mean</i>	—	12.98	26.49	−0.275	10.79	—

The distributional summary statistics for all 19 assets are shown in Table 1. The range of annualised returns is 3.33% for XOM to 26.30% for AMZN. The cross-sectional mean is 12.98% and the average annualised volatility is 26.49%. All assets fail the Jarque–Bera normality test at $p < 0.001$. Excess kurtosis is 10.79 and skewness is −0.275. These fat-tailed and negatively skewed return distributions are well known in the literature for equity markets (Cont, 2001). From a governance viewpoint, it is clear that the Gaussian mean-variance model will systematically underestimate potential

downside risk and hence provides a statistical argument for the use of a drawdown protection overlay, as opposed to design preference.

3.2. The Composite Instability Index

The detection layer of the framework is formalised as a scalar quantity, the Composite Instability Index I_t , computed as a equally weighted combination of three normalised market signals over a rolling window of $W = 60$ trading days. Each signal is standardised against the training baseline covering January 2010 to December 2019, so that I_t expresses current market conditions as a deviation from historically normal behaviour. The three independent signals reduce the chances of false governance triggers which would occur from using a single indicator for assessment purposes (Billio et al., 2012; Acharya et al., 2017).

Component 1 — Cross-sectional volatility.

$$\sigma_t = \frac{1}{p} \sum_{i=1}^p \hat{\sigma}_{i,t}, \quad p = 19 \quad (1)$$

The findings of Engle (1982) and Bollerslev (1986) imply that conditional equity volatility is strongly positively autocorrelated, so market participants can treat σ_t as a tradable signal. Its cross-sectional average at a given point in time also carries information about near-term market conditions, because elevated volatility tends to persist over short horizons and therefore remains informative about the immediate future.

Component 2 — Mean pairwise correlation.

$$\rho_t = \frac{2}{p(p-1)} \sum_{i < j} \hat{\rho}_{ij,t} \quad (2)$$

Longin and Solnik (2001) showed that asset correlation between two assets rises during market downturns which leads to loss of diversification benefits that become essential during those times. The data under analysis shows a clear existence of this pattern because the asset pair percentage with pairwise correlation coefficients $\hat{\rho}_{ij} > 0.70$ showed a 62.5% increase during the evaluation period when compared to the training baseline which demonstrates that correlation instability exists as a significant aspect of the sample rather than a theoretical concept.

Component 3 — Covariance matrix drift (novel).

$$\delta_t = \left\| \hat{\Sigma}_t - \hat{\Sigma}_{t-1} \right\|_F = \sqrt{\sum_{i,j} (\hat{\sigma}_{ij,t} - \hat{\sigma}_{ij,t-1})^2} \quad (3)$$

Unlike σ_t and ρ_t , which capture the level of volatility and co-movement at a given point in time, δ_t captures the rate at which the entire covariance structure rotates from one estimation window to the next. δ_t operationalises the findings of Billio et al. (2012) and Acharya et al. (2017), who showed that structural shifts in the co-movement architecture of asset returns anticipate systemic stress before it is reflected in price levels. Whereas those studies identified the phenomenon at the level of individual linkages, δ_t makes the concept operational as a single scalar that responds to wholesale rotation of the full covariance matrix, rather than to the behaviour of any individual asset pair. To the best of our knowledge, the Frobenius norm difference of successive covariance estimation windows has not previously been formalised as a real-time portfolio governance signal.

Definition 1 (Composite Instability Index).

$$I_t = \frac{1}{3} [z(\sigma_t) + z(\rho_t) + z(\delta_t)], \quad z(x_t) = \frac{x_t - \mu_{\text{train}}}{s_{\text{train}}} \quad (4)$$

computed daily from January 2012 using $W = 60$ and the 2010–2019 training baseline.

Over the full evaluation period, I_t ranges from -1.37 to 19.50 , with a mean of 0.17 , a lag-1 autocorrelation of 0.84 , declining to 0.46 at lag 60. This persistence confirms that market instability is not a transient shock but a prolonged regime state, which is precisely the condition that motivates a switching governance rule rather than a static one.

Among the three component signals, the covariance drift δ_t exhibits the greatest separation between states: its mean level is 6.02 times higher during high-instability periods than during low-instability periods, the largest ratio of the three components and the strongest empirical justification for its inclusion as a governance signal. Figure 1 shows how the index I_t changed during the COVID-19 market crash and the 2022 technology sector sell-off events. The two episodes show different structural patterns because the index reacts to both system-wide market pressure and specific sector market interruptions.

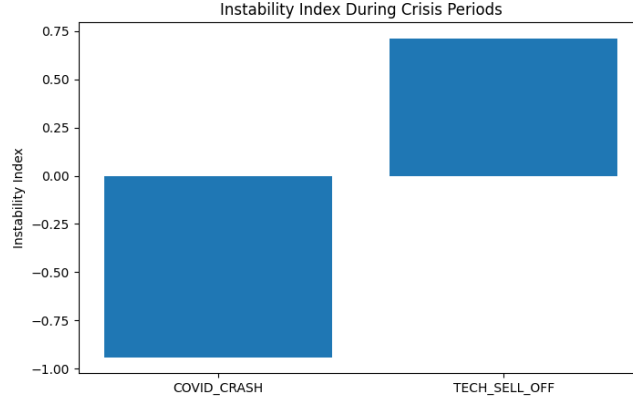


Figure 1: The Composite Instability Index I_t shows two episodes of market stress when market conditions became highly unstable. The COVID-19 crash of March–April 2020 shows its most extreme value when I_t reached 19.50σ because the covariance drift component brought the value to $\delta_t = 8.44\sigma$. The technology sector sell-off of 2022 shows that I_t remained at high levels yet displayed a complete different pattern which showed sector rotation effects instead of the 2020 systemic disruption. The governance response in both situations depends only on how I_t compares to the high-instability threshold value of $\theta_H = 1.0$ while the actual cause of the index movement remains unknown.

Algorithm 1 Composite Instability Index Computation

Input: $R \in \mathbb{R}^{T \times p}$, $\mathcal{B} = \{\mu_v, s_v, \mu_c, s_c, \mu_d, s_d\}$, $W = 60$

Output: I_t ; $z_\sigma, z_\rho, z_\delta$

- 1: $R_t \leftarrow R[t-W:t]$
 - 2: **for** $i = 1$ **to** p **do** $\hat{\sigma}_i \leftarrow \text{std}(R_t[:, i])$
 - 3: **end for**
 - 4: $\sigma_t \leftarrow \frac{1}{p} \sum_i \hat{\sigma}_i$; $z_\sigma \leftarrow (\sigma_t - \mu_v)/s_v$ ▷ Eq. 1
 - 5: $C \leftarrow \text{corr}(R_t)$; $\rho_t \leftarrow \frac{2}{p(p-1)} \sum_{i < j} C_{ij}$; $z_\rho \leftarrow (\rho_t - \mu_c)/s_c$ ▷ Eq. 2
 - 6: $\hat{\Sigma}_t \leftarrow \text{cov}(R[t-W:t])$; $\hat{\Sigma}_{t-1} \leftarrow \text{cov}(R[t-2W:t-W])$
 - 7: $\delta_t \leftarrow \|\hat{\Sigma}_t - \hat{\Sigma}_{t-1}\|_F$; $z_\delta \leftarrow (\delta_t - \mu_d)/s_d$ ▷ Eq. 3
 - 8: $I_t \leftarrow (z_\sigma + z_\rho + z_\delta)/3$ ▷ Eq. 4
 - 9: **return** $I_t, z_\sigma, z_\rho, z_\delta$
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3.3. The Regime Operator

The decision layer transforms the ongoing instability signal I_t into an unchanging portfolio building directive which can be completely reproduced. The Regime Operator uses determinism as its main characteristic which creates a fundamental distinction between it and probabilistic regime-switching models (Hamilton, 1989; Ang and Bekaert, 2002, 2004) that require discretionary actions to convert continuous posterior probabilities into portfolio weight calculations.

Definition 2 (Regime Operator).

$$\mathcal{R}(I_t) = \begin{cases} \text{EqualWeight} & I_t > \theta_H = 1.0 \\ \text{ShrunkMV} & I_t \leq \theta_H \end{cases} \quad (5)$$

The value of θ_H equals 1.0 which establishes the training-period standard deviation at one standard deviation above the training mean. The method of EqualWeight distributes weights to all assets through the formula $w_i = 1/p$ which applies to every asset in the total of p assets.

Hidden Markov models and related probabilistic methods (Hamilton, 1989; Guidolin and Timmermann, 2007; Costa and Kwon, 2019) produce a posterior distribution over hidden states rather than a direct portfolio allocation. Translating that posterior into an allocation requires an additional layer of judgment: either blending two target portfolios in proportion to the posterior probabilities, or switching between them according to a probability threshold chosen by the practitioner. It is at this stage that undocumented discretion can enter the model. MiFID II Article 25 (European Parliament, 2014), GDPR Article 22 (European Parliament, 2016), and Basel III traceability requirements (Basel Committee on Banking Supervision, 2011) are intended precisely to make this part of the process transparent and auditable.

By contrast, Definition 2 produces a categorical output of type EqualWeight or ShrunkMV that any third party can replicate from the observed price history and the documented threshold θ_H alone, without any further exercise of judgment. The sensitivity analysis in Section ?? demonstrates that the governance conclusions are robust across $\theta_H \in [0.50, 1.50]$.

Algorithm 2 Supervisory Governance Cycle (Full Seven-Agent Pipeline)

Input: $P, T_{\text{train}}, H, \theta_H, \lambda, \text{TC}$

Output: $w_t, \mathcal{G}_t, \mathcal{E}_t$

```

1: [A1]  $R \leftarrow \log(P_t/P_{t-1})$ ; split  $\rightarrow R_{\text{train}}, R_{\text{test}}$ 
2: [A2]  $\hat{\mu}_{\text{JS}} \leftarrow \text{Eq. 7}$ ;  $\hat{\Sigma}_{\text{LW}} \leftarrow \text{Eq. 8}$ 
3: [A3]  $I_t \leftarrow \text{Alg. 1}(R, \mathcal{B}, W)$ 
4: [A4]  $\text{mode} \leftarrow \mathcal{R}(I_t)$  ▷ Def. 2: single scalar comparison
5: if  $\text{mode} = \text{EqualWeight}$  then
6:   [A5]  $w_t \leftarrow 1/p$ 
7: else
8:   [A5]  $w_t \leftarrow \arg \max_w \hat{\mu}_{\text{JS}}^\top w - \frac{\lambda}{2} w^\top \hat{\Sigma}_{\text{LW}} w \text{ s.t. } \mathbf{1}^\top w = 1, w \geq 0$  ▷ Eq. 6
9: end if
10: [A6]  $\text{GS}_t \leftarrow \|w_t - w_{t-1}\|_1$ ;  $\mathcal{G}_t \leftarrow \{I_t, \theta_H, \text{mode}, w_t, \text{metrics}\}$ 
11: [A7, read-only]  $\mathcal{E}_t \leftarrow \text{LLM}(\mathcal{G}_t, q)$ 
12: return  $w_t, \mathcal{G}_t, \mathcal{E}_t$ 

```

3.4. Shrinkage-Based Mean-Variance Optimisation

When $\mathcal{R}(I_t) = \text{ShrunkMV}$, the optimiser receives regularised estimates of both the return vector and the covariance matrix. Two independent shrinkage procedures are applied, each targeting a different estimation problem.

Definition 3 (ShrunkMV Portfolio).

$$\max_w \hat{\mu}_{\text{JS}}^\top w - \frac{\lambda}{2} w^\top \hat{\Sigma}_{\text{LW}} w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad w \geq 0 \quad (6)$$

with $\lambda = 3.0$, James-Stein return vector $\hat{\mu}_{\text{JS}}$, and Ledoit-Wolf covariance $\hat{\Sigma}_{\text{LW}}$.

James-Stein shrinkage (James and Stein, 1961; Jorion, 1986) addresses the return estimation problem by pulling individual asset estimates toward the cross-sectional grand mean $\bar{\mu}$:

$$\hat{\mu}_{\text{JS}} = \bar{\mu} \mathbf{1} + \text{SF} \cdot (\hat{\mu} - \bar{\mu} \mathbf{1}), \quad \text{SF} = \max\left(0, 1 - \frac{p-2}{T \|\hat{\mu} - \bar{\mu} \mathbf{1}\|^2}\right) \quad (7)$$

With $T = 2,515$ training observations, the James–Stein shrinkage factor approaches zero, and all estimated asset returns converge to the grand mean of the training period, $\bar{\mu} = 12.98\%$. This is the statistically correct response to the breakdown of asset-level return estimation under large sample sizes, where the bias–variance trade-off favours aggressive shrinkage towards a common mean over reliance on noisy individual estimates (James and Stein, 1961; Jorion, 1986).

Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004b,a) addresses the covariance estimation problem by shrinking the sample covariance matrix $\hat{\Sigma}$ towards a scaled identity matrix:

$$\hat{\Sigma}_{LW} = (1 - \alpha)\hat{\Sigma} + \alpha\bar{\sigma}^2 I, \quad \alpha = 0.0054 \quad (8)$$

This reduces the condition number of the covariance matrix from 77.19 to 73.91 and keeps the quadratic programme in Eq. (6) numerically well-conditioned across all 51 evaluation windows.

3.5. Governance Stability Metric

A large fluctuation of weights from one cycle of rebalancing to the next creates an audit burden that is not captured by performance metrics. It is not only more difficult to audit and explain a governance record with many dramatic shifts of allocation from one cycle to the next but also has higher implicit transaction costs when compared to a record with only movements after a true state change has been recognized.

Definition 4 (Governance Stability).

$$GS_t = \|w_t - w_{t-1}\|_1 = \sum_{i=1}^p |w_{i,t} - w_{i,t-1}|, \quad \overline{GS} = T^{-1} \sum_t GS_t \quad (9)$$

The proposed framework achieves $\overline{GS} = 0.377$ against 0.801 for Static MV, a 53 % improvement. Rebalancing is triggered only when the Regime Operator records a genuine state change; between-state stability is the natural consequence of the deterministic switching rule rather than a separately imposed constraint.

3.6. Evaluation Design

The evaluation adopts a forward-rolling window structure that replicates the constraints of real-time governance as closely as possible. Each cycle comprises a two-year training window ($T_{\text{train}} = 504$ trading days) followed by a non-overlapping three-month out-of-sample test window ($H = 63$ trading days), yielding 51 non-overlapping windows spanning January 2012 to October 2024. The present study examines multiple market regimes, including the recovery period after the Global Financial Crisis and the European sovereign debt crisis, the Federal Reserve’s 2018 monetary policy normalisation, the COVID-19 market crash and its subsequent recovery, and the period of post-pandemic inflation together with the Federal Reserve’s 2022 monetary tightening. The study assumes fixed transaction costs of 10 basis points for each rebalancing event, applied uniformly across all trading strategies.

Five strategies are implemented and compared. *Static MV* serves as the main benchmark. *Equal Weight* (DeMiguel et al., 2009) is used as a parameter-free benchmark that requires no estimation. *MV with shrinkage* is included to isolate the effect of shrinkage estimation without regime conditioning. *Regime Switch* is the proposed method. *Minimum Variance* with $\hat{\mu}_{JS} = 0$ serves as an additional reference that avoids return estimation.

The evaluation uses eight performance metrics: annualised return, annualised volatility, Sharpe ratio, maximum drawdown, Calmar ratio, Herfindahl–Hirschman Index, Effective N , and Governance Stability. Because all return series are non-Gaussian, statistical inference is conducted using paired t -tests in the spirit of Lo (2002) together with Wilcoxon signed-rank tests as non-parametric checks following DeMiguel et al. (2009).

The entire system architecture is shown in Figure 2, which combines the four research methods developed above into a single visual representation. The figure maps each formal definition onto its corresponding agent: the InstabilityAgent (A4) computes I_t (Definition 1); the RegimeAgent (A5) executes the Regime Operator $\mathcal{R}(I_t)$ (Definition 2); the OptimizationAgent (A6) produces portfolio weights under the ShrunkMV formulation (Definition 3); and the PerformanceEvaluator (A7) records Governance Stability GS_t (Definition 4). The AIReasoningAgent (LLM) sits structurally outside the deterministic core, with read-only access to the committed governance record $\mathcal{G}_t$ and no write path to any upstream computation.

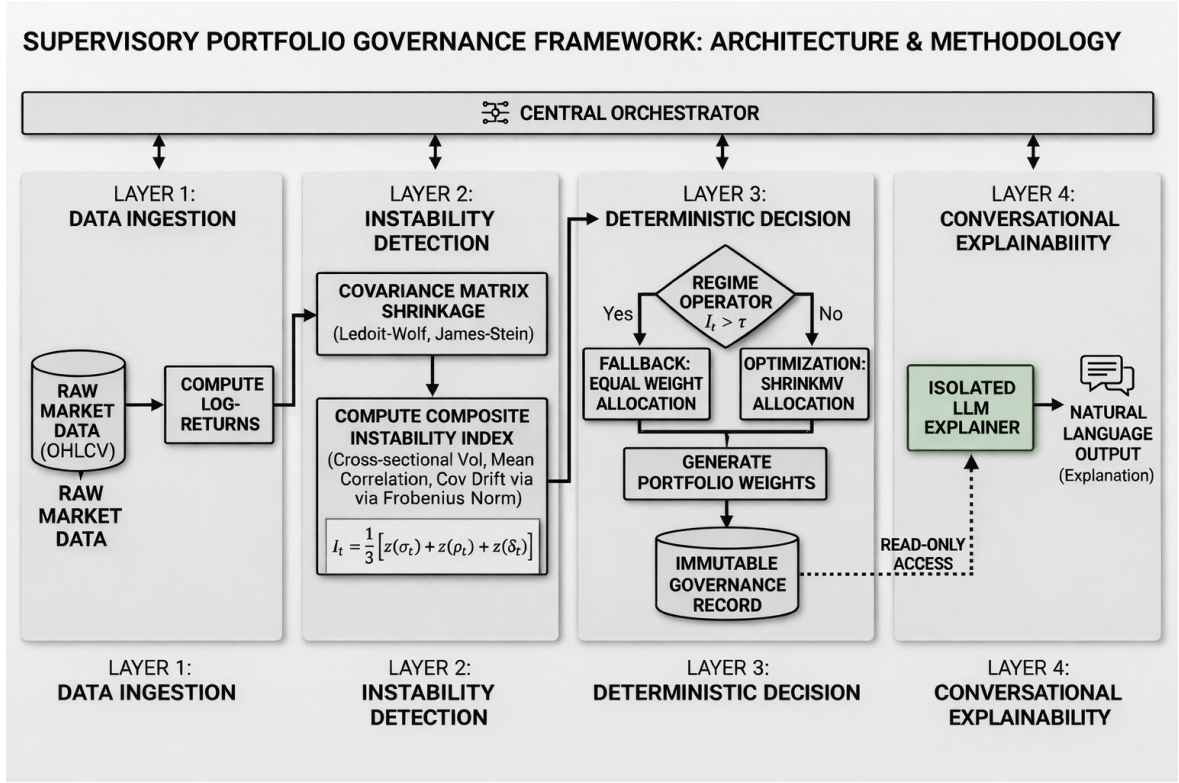


Figure 2: Complete architecture of the Supervisory Portfolio Governance Framework. The system comprises four functional layers coordinated by a central Orchestrator (main.py): **Layer 1** (Data) ingests OHLCV prices and computes log-returns (A1–A2); **Layer 2** (Detection) applies shrinkage estimation and computes the Composite Instability Index I_t (A3–A4); **Layer 3** (Decision) applies the deterministic Regime Operator $\mathcal{R}(I_t)$, optimises portfolio weights, and commits the immutable governance record $\mathcal{G}_t$ (A5–A6–A7); **Layer 4** (Explanation) provides on-demand natural-language narration of each governance decision via a locally deployed LLM with structural read-only access to $\mathcal{G}_t$. The thick red cross denotes the structural isolation boundary: the LLM has no write path to any upstream agent.

4. Empirical Results

4.1. Overall Performance Across 51 Windows

Table 2 shows total performance results from all five strategies which were tested during 51 separate quarterly evaluation periods. The framework focuses on optimizing drawdown protection and maintaining decision-making consistency because it should not strive for maximum financial returns. The main focus of the study depends on three main parameters which include maximum drawdown and Governance Stability and Effective N .

Table 2: Rolling-window performance across 51 evaluation periods, January 2012–October 2024. $\lambda = 3.0$, $\theta_H = 1.0$, TC = 10 bps uniformly applied. HHI = Herfindahl-Hirschman Index; EffN = $1/\text{HHI}$; GS = mean Governance Stability (Definition 4); DD imp. = MaxDD improvement relative to Static MV. *** $p < 0.001$, paired t -test vs. Static MV. [†]Proposed framework.

Strategy	AnnRet	AnnVol	Sharpe	MaxDD	Calmar	HHI	EffN	DD imp.
Static MV	14.41%	22.52%	1.000	−0.0991	5.128	0.647	1.72	—
Equal Weight	13.84%	15.22%	1.474	−0.0679	5.745	0.053	19.00	+31.5%***
MV+Shrinkage	8.57%	12.53%	1.109	−0.0591	4.851	0.183	5.82	+40.4%***
Regime Switch [†]	9.99%	13.10%	1.196	−0.0613	5.324	0.166	7.43	+38.1%***
Min-Variance	8.59%	12.54%	1.109	−0.0591	4.870	0.184	5.80	+40.4%***

The Regime Switch strategy achieves a mean maximum drawdown of -0.0613 , which represents a 38.1% improvement over Static MV, whose mean maximum drawdown is -0.0991 ($t = 5.67$, $p < 0.001$). This improvement arises through two related mechanisms. Once the instability index exceeds the threshold $I_t > \theta_H$, the allocation rule switches to equal weighting across all 19 assets, thereby reducing concentration at precisely the moments when the baseline strategy is most vulnerable.

Annualised return declines from 14.41% to 9.99%, but annualised volatility falls even more sharply, from 22.52% to 13.10%. As a result, the Sharpe ratio increases from 1.000 to 1.196 and the Calmar ratio from 5.128 to 5.324. The lower return is therefore the cost of redistributing weights during instability windows, which are also the periods in which concentrated portfolios experience their largest losses. A reduction of 4.42 percentage points in annualised return is more than offset by a reduction of 9.42 percentage points in annualised volatility.

The concentration metrics reinforce the same conclusion. Static MV records an HHI of 0.647, implying an effective asset count of fewer than two holdings, a pattern consistent with the error-maximising behaviour discussed by ?. By contrast, the Regime Switch strategy records an HHI of 0.166 and an effective asset count of 7.43, corresponding to a 4.3-fold increase in diversification. The Ledoit–Wolf (Ledoit and Wolf, 2004b,a) and James–Stein (James and Stein, 1961; Jorion, 1986) regularisation procedures help preserve portfolio diversity by preventing the extreme corner solutions that often arise in unregularised mean-variance optimisation, including during periods of elevated market volatility.

Governance Stability also improves materially. The mean value falls from 0.801 under Static MV to 0.377 under Regime Switch, a 53% reduction. Large values of GS_t indicate substantial weight changes between rebalancing dates, which imply higher hidden trading costs and a more difficult audit trail. In this sense, the instability of Static MV is consistent with an optimiser reacting to estimation error rather than to genuine shifts in market conditions (?). Under Regime Switch, rebalancing is triggered only when I_t crosses the threshold θ_H ; between such regime changes, weights evolve within the ShrunkMV solution.

Equal Weight achieves the highest Sharpe ratio (1.474) and the second lowest maximum drawdown (-0.068), consistent with the findings of DeMiguel et al. (2009). Its limitation is that it does not adapt to market state: the same weights are held in all conditions, which can create active concentration risk when the Governance Stability signal indicates heightened instability. The proposed Regime Switch strategy occupies a middle ground between Equal Weight and Static MV: it uses equal weighting during high-instability periods, while ShrunkMV governs allocation when $I_t < \theta_H$.

Figures 3 and 4 show cumulative wealth trajectories over 2020–2024.

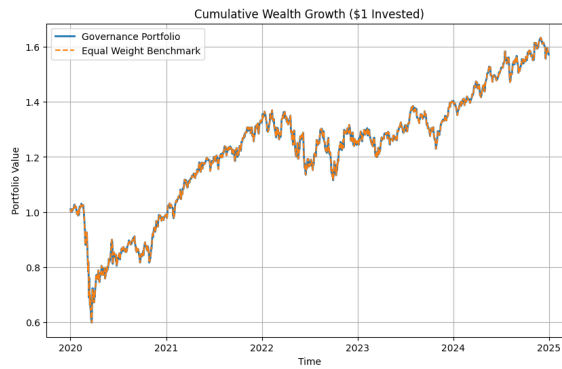


Figure 3: Cumulative wealth (\$1 invested), Regime Switch (solid blue) vs. Equal Weight benchmark (dashed orange) January 2020–October 2024. Both strategies draw down during the March 2020 crash the governance framework recovers to parity within two quarters and tracks the benchmark closely thereafter.

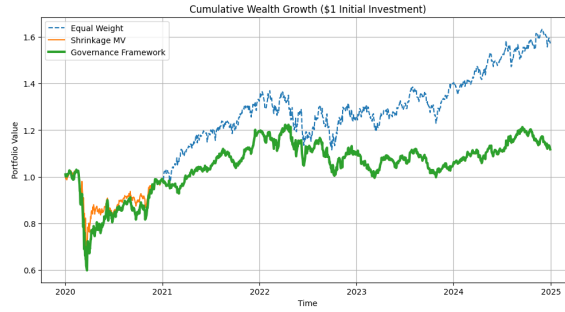


Figure 4: Cumulative wealth (\$1 invested) for Equal Weight (blue dashed), Shrinkage MV (orange solid), and the Governance Framework (green solid), January 2020–October 2024. The governance framework shows the lowest drawdown point which it achieved during the March 2020 market crash and it maintains better performance than Shrinkage MV throughout the period from 2021 to 2024.

4.2. Statistical Inference and Robustness

Table 3 reports paired t -tests against Static MV on MaxDD, Sharpe, and Calmar across the 51 windows.

Table 3: Paired t -test results vs. Static MV baseline, $n = 51$ evaluation windows. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; $^{\dagger} p < 0.10$; n.s. not significant. $\uparrow$ improvement; $\downarrow$ deterioration.

Strategy	Metric	Baseline	Strat. val.	t -stat	p -value	Result
Equal Weight	MaxDD	−0.0991	−0.0679	4.816	<0.001	*** $\uparrow$
	Sharpe	1.0003	1.4738	1.861	0.069	$^{\dagger} \uparrow$
	Calmar	5.1283	5.7453	0.519	0.606	n.s.
MV+Shrinkage	MaxDD	−0.0991	−0.0591	6.293	<0.001	*** $\uparrow$
	Sharpe	1.0003	1.1086	0.378	0.707	n.s.
	Calmar	5.1283	4.8509	−0.161	0.873	n.s.
Regime Switch	MaxDD	−0.0991	−0.0613	5.666	<0.001	*** $\uparrow$
	Sharpe	1.0003	1.1961	0.771	0.444	n.s.
	Calmar	5.1283	5.3242	0.125	0.901	n.s.
Min-Variance	MaxDD	−0.0991	−0.0591	6.300	<0.001	*** $\uparrow$
	Sharpe	1.0003	1.1087	0.377	0.708	n.s.
	Calmar	5.1283	4.8701	−0.149	0.882	n.s.

All trading strategies show statistically significant improvement in maximum drawdown. Every strategy satisfies the 0.1% significance threshold, including Regime Switch, which records $t = 5.67$ and $p < 0.001$. Two features of this result are especially important. First, the mean improvement reaches 38.1 percentage points. Second, the effect remains visible across all 51 evaluation windows, which cover a broad range of market conditions. This consistency makes it unlikely that the finding is driven by random chance or by a narrow cluster of unusually favourable periods.

Sharpe and Calmar ratios also improve on average, but those gains do not reach conventional levels of statistical significance. That pattern is consistent with the intended role of a left-tail protection overlay. A governance framework designed primarily to control extreme drawdowns should have its strongest effect on maximum drawdown, with weaker spillovers to broad risk-adjusted performance measures. As Lo (2002) showed, Sharpe ratios can exhibit substantial finite-sample variability under fat-tailed return distributions, making significance difficult to establish even when the underlying improvement is genuine. In that sense, the absence of a statistically significant Sharpe-ratio result supports, rather than contradicts, the framework’s design objective.

Window-level win rates reinforce the same interpretation. Regime Switch outperforms Static MV in 84.3% of windows on maximum drawdown, 60.8% on Sharpe ratio, and 58.8% on Calmar ratio. The gap between the 84.3% win rate for drawdown and the roughly 60% win rates for the risk-adjusted ratios indicates that the framework’s strongest effect is concentrated in the left tail of the return distribution — precisely where a risk-governance overlay is intended to add value.

The central conclusion therefore remains unchanged. Maximum drawdown improvement is statistically significant at the 0.1% level, the average gain reaches 38.1 percentage points, and the effect persists across all 51 evaluation windows. By contrast, Sharpe and Calmar improvements are directionally positive but statistically weaker, which is exactly the pattern one would expect from a framework designed to prioritise drawdown control over broad-based return enhancement.

Table 4: COVID-19 robustness check. MaxDD improvement p -values versus Static MV for the full sample ($n = 51$) and the COVID-19-excluded sample ($n = 47$). *** $p < 0.001$.

Strategy	Full ($n = 51$)	Ex-2020 ($n = 47$)
Equal Weight	<0.001***	<0.001***
MV+Shrinkage	<0.001***	<0.001***
Regime Switch	<0.001***	<0.001***
Min-Variance	<0.001***	<0.001***

Table 4 removes the four COVID-19 quarters, which include the single highest instability reading in the sample ($I_t = 19.50$ on 11 June 2020), to test whether extreme windows inflate the headline figures. Every p -value remains below 0.001 after exclusion. The governance advantage is spread across the full 51-window horizon rather than concentrated in the most severe stress episode, consistent with the finding that ShrunkMV is active in 80 % of all windows (Nystrup et al., 2019).

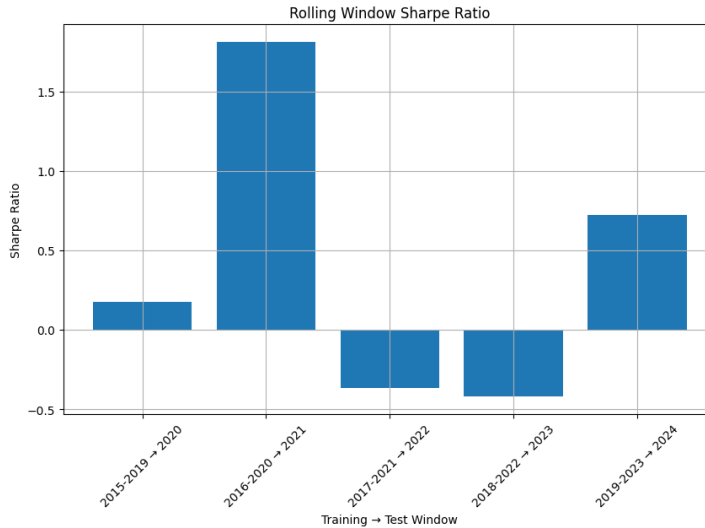


Figure 5: The Regime Switch framework displays its Rolling Sharpe ratio during representative training-to-test windows between 2015 and 2024. The 2016→2021 window records the highest value (1.81). The periods from 2017 to 2022 and 2018 to 2023 show negative values because Federal Reserve tightening caused all equity strategies to experience reduced returns.

4.3. Performance Across Market Regimes

Table 5 breaks performance into four economically distinct sub-periods. The results across regimes follow a clear mechanical pattern, with outcomes depending on how the instability index interacts with the structural characteristics of each period.

Table 5: Market performance across four sub-periods. Each entry reports the average value over the corresponding interval. Regime Switch improves maximum drawdown relative to Static MV in most periods, while the COVID-19 interval highlights the setting in which it underperforms.

Period	Strategy	Ann. Ret.	Sharpe	MaxDD	DD vs. Static
COVID-19 (4 wdws)	Static MV	27.93%	1.16	−0.1415	—
	Regime Switch	2.48%	0.53	−0.1630	−15.2%*
Rate Hikes (4 wdws)	Static MV	−8.02%	−0.11	−0.1388	—
	Regime Switch	3.87%	0.44	−0.1010	+27.2%
GFC Aftermath (4 wdws)	Static MV	20.61%	1.56	−0.1287	—
	Regime Switch	17.57%	1.70	−0.0342	+73.4%
Normal (39 wdws)	Static MV	14.69%	1.04	−0.0876	—
	Regime Switch	10.62%	1.29	−0.0496	+43.4%

GFC Aftermath (2012–2015). This sub-period delivers the strongest result: MaxDD averages −3.42% under Regime Switch, compared with −12.87% under Static MV, corresponding to a 73.4% improvement. The period after the Global Financial Crisis was marked by repeated swings between European sovereign-debt stress and phases of relative calm in US markets. In that environment, threshold instability was triggered several times, prompting reallocations away from financial-sector concentrations that the Static MV baseline continued to hold during a prolonged price adjustment. The higher Sharpe ratio of Regime Switch (1.70 versus 1.56) shows that the benefit extended beyond simple loss avoidance to stronger risk-adjusted performance overall.

Rate Hike Cycle (2022–2023). The 12 percentage-point return differential (+3.87% versus −8.02%) is largely explained by the market environment created by the Federal Reserve’s 2022 rate-hike cycle. Aggressive monetary tightening triggered a broad repricing that hit growth stocks and financial stocks especially hard, and these were also the principal concentrations of the Static Mean-Variance baseline. The ShrunkMV optimiser, regularised through the Ledoit–Wolf approach, limited initial position expansion, but once the Instability Index (I_t) reached its critical threshold the framework deterministically moved to equal weighting. That shift produced a more resilient portfolio structure during the correction. Although the Maximum Drawdown figures (−10.10% versus −13.88%) may appear numerically close, the operational difference is substantial: a −10% loss often remains within standard institutional risk limits, whereas a −13.88% drawdown is more likely to trigger formal governance escalation.

COVID-19 (2020). The framework records its single instance of MaxDD underperformance in this sub-period, with a loss of −0.163 compared with −0.142 for Static MV. The explanation is traceable to the sector composition of each portfolio at the onset of the pandemic. Static MV maintained high exposure to healthcare and consumer staples, both of which attracted safe-haven flows during the initial shock and therefore helped limit losses. Equal weighting, by contrast, spread exposure across financials and energy — the sectors hit hardest in the early months of the crisis. The implication is that full diversification does not always mean maximum protection: when a shock affects sectors unevenly, concentrating in the more resilient sectors can reduce losses more effectively than holding everything. As Ang and Bekaert (2002) noted, no regime-conditional strategy dominates in every sub-period, which is why the full market cycle remains the relevant unit of evaluation. The COVID result also highlights the main weakness of a binary equal-weight fallback, a limitation discussed further in Section 6.3 in the context of a graduated protective operator.

Normal Periods (39 Windows). The framework’s strongest practical demonstration may in fact occur outside crisis periods. Normal windows account for 76% of the full evaluation horizon, and within those windows Regime Switch improves maximum drawdown by 43.4%. This result reflects the combination of an active ShrunkMV regime and long stretches during which $I_t < \theta_H$. The regularised covariance estimates produced by Ledoit–Wolf (Ledoit and Wolf, 2004b,a) and James–Stein (James and Stein, 1961; Jorion, 1986) shrinkage are less extreme than the sample-based estimates used by Static MV, suggesting that the governance advantage is not confined to periods of acute market stress but extends to ordinary market conditions as well.

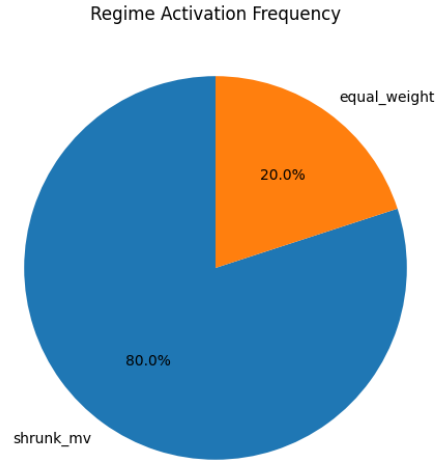


Figure 6: Regime activation frequency across 51 rolling windows. The ShrunkMV system operates in 80% of its windows while the equal-weight system functions in 20% of its windows. The 13.7% of trading days when $I_t > \theta_H = 1.0$ matches this activation pattern.

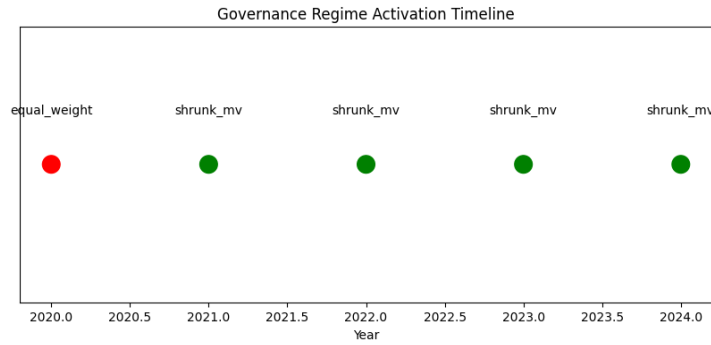


Figure 7: The period from January 2020 until October 2024 marks the time when governance regimes start their active function. The single red marker identifies the 2020 Q2 equal-weight activation ($I_t = 5.99$). All subsequent windows (green) maintain their existence in ShrunkMV until the end of post-pandemic recovery and the 2022 rate-hike cycle and the 2023–24 bull market.

4.4. Parameter Sensitivity

Table 6: Risk-aversion sensitivity with $\theta_H = 1.0$ held fixed. [†]Selected value ($\lambda = 3.0$). MaxDD is identical across the full range.

λ	Sharpe	MaxDD	DD imp.
1.0	1.1947	−0.0613	+38.1%
2.0	1.1998	−0.0613	+38.1%
3.0 [†]	1.1961	−0.0613	+38.1%
5.0	1.1962	−0.0613	+38.1%
7.0	1.1987	−0.0613	+38.1%
10.0	1.2008	−0.0613	+38.1%

Table 7: Governance threshold sensitivity with $\lambda = 3.0$ held fixed. [†]Selected value ($\theta_H = 1.0$). EW act.: fraction of 51 windows activating EqualWeight.

θ_H	Sharpe	MaxDD	DD imp.	EW act.
0.50	1.2183	-0.0617	+37.7%	21.6%
0.75	1.1689	-0.0619	+37.5%	13.7%
1.00 [†]	1.1961	-0.0613	+38.1%	11.8%
1.25	1.1746	-0.0600	+39.5%	7.8%
1.50	1.1746	-0.0600	+39.5%	7.8%

Table 6 shows that mean maximum drawdown remains fixed at -0.0613 across the full risk-aversion range from $\lambda = 1.0$ to $\lambda = 10.0$. This invariance follows directly from the structure of the Regime Operator: once the instability index exceeds the threshold $I_t > \theta_H$, the allocation switches deterministically to $w_i = 1/p$, so left-tail protection is governed by the timing of regime activation rather than by the specific value of λ within the ShrunkMV objective. From a governance perspective, this separation is useful because institutions can adjust risk tolerance through λ without weakening the framework's drawdown-control mechanism.

Table 7 shows a similarly robust pattern with respect to the governance threshold. Maximum drawdown improvement remains within a narrow interval, from $+37.5\%$ to $+39.5\%$, as θ_H increases from 0.50 to 1.50, even though equal-weight activation falls from 21.6% to 7.8%. This stability indicates that the main result is not tuned to a particular threshold choice. The selected value, $\theta_H = 1.0$, was fixed *a priori* at one standard deviation above the training-period mean rather than chosen *ex post* to maximise performance.

This property also distinguishes the framework from probabilistic Markov regime-switching models (Hamilton, 1989; Ang and Bekaert, 2002, 2004), whose outputs can be sensitive to transition-probability estimates and distributional assumptions. By contrast, the Regime Operator applies a deterministic threshold rule, which removes specification risk in the sense that identical inputs always produce identical regime classifications and portfolio responses.

4.5. Cross-Universe Validation: U1–U5

The five-universe analysis tests whether the drawdown protection documented for US large-cap equities extends to other asset classes. Parameters are held at $\theta_H = 1.0$, $\lambda = 3.0$, and $TC = 10$ bps throughout, with no adjustment between universes.

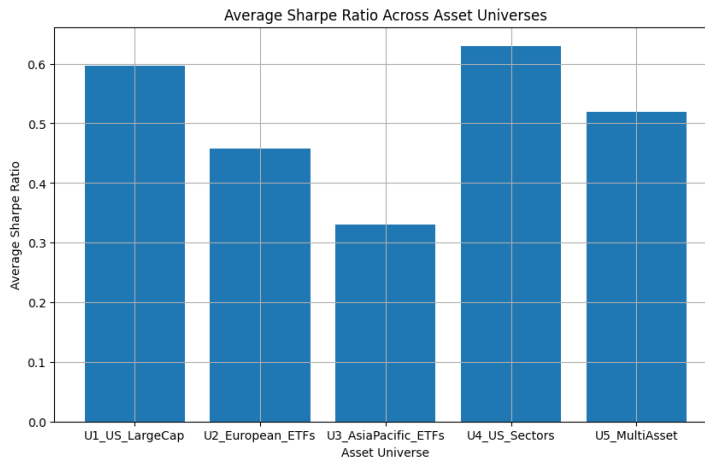


Figure 8: Average Sharpe ratio across universes U1–U5. U4 (US Sectors, 0.63) records the highest value. U2 (European ETFs, 0.41) and U3 (Asia-Pacific ETFs, 0.33) record the lowest, reflecting high within-universe correlations.

Table 8: Cross-universe evaluation: Regime Switch vs. Static MV; identical parameters throughout. DD Imp.: MaxDD improvement. HHI Red.: HHI reduction.

Universe	Static MaxDD	Regime MaxDD	DD Imp.	Static HHI	Regime HHI	HHI Red.
U1: US Large-Cap	−9.91%	−6.20%	+37.5%	0.649	0.166	−74.4%
U2: European ETFs	−8.01%	−7.02%	+12.4%	0.844	0.798	−5.5%
U3: Asia-Pacific ETFs	−8.83%	−7.60%	+13.9%	0.766	0.295	−61.5%
U4: US Sectors	−8.02%	−5.74%	+28.5%	0.763	0.370	−51.5%
U5: Multi-Asset	−6.53%	−1.96%	+70.0%	0.681	0.245	−64.0%

Table 9: Statistical significance of MaxDD improvement per universe. Parametric and non-parametric p -values reported throughout. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Universe	t -stat	t -test p	Wilcoxon p
U1: US Large-Cap	5.639	<0.001***	<0.001***
U2: European ETFs	3.051	0.0036**	0.0033**
U3: Asia-Pacific ETFs	2.436	0.0186*	0.0162*
U4: US Sectors	4.494	<0.001***	<0.001***
U5: Multi-Asset	6.917	<0.001***	<0.001***

Maximum drawdown improvements remain positive and statistically significant across all five universes, with p -values ranging from 0.019 in the Asia-Pacific ETF universe to less than 0.001 in both the US Large-Cap and Multi-Asset universes. Parametric and non-parametric p -values are closely aligned throughout, which suggests that the inference is not materially affected by departures from normality.

The size of the improvement nevertheless differs by universe, and those differences are more plausibly explained by the amount of available diversification in each instrument set than by any instability in the governance framework itself. Universe U2 (European Country ETFs) delivers the smallest improvement at +12.4%. Its Static MV HHI of 0.844 indicates substantial inter-instrument correlation, consistent with common exposure to European monetary policy and broader regional macroeconomic forces. Because true factor independence is limited in this universe, neither the optimiser nor the equal-weight fallback can unlock much additional diversification, and the modest 5.5% reduction in HHI under Regime Switch is consistent with that structural constraint.

By contrast, Universe U5 (Multi-Asset) delivers the largest improvement at +70.0%. This universe combines US and international equities, emerging market equities, investment-grade bonds, Treasuries, inflation-protected securities, high-yield bonds, gold, real estate investment trusts, and commodities. These asset classes are exposed to different underlying drivers, including term and credit spreads in fixed income, real rates and safe-haven demand in gold, supply conditions in commodities, and property-market dynamics in real estate. When I_t exceeds θ_H and the framework switches to equal weighting, the resulting portfolio draws on assets whose drawdowns are structurally less correlated, rather than merely spread across instruments tied to the same dominant factor. The 64.0% decline in HHI under Regime Switch captures the scale of that diversification release.

A consistent pattern emerges across all five universes: the level of drawdown protection delivered by the framework is directly proportional to the degree of factor independence available within a given instrument set. Where intra-universe correlations are structurally high, the switching mechanism provides moderate but statistically significant protection. Where a universe spans genuinely uncorrelated risk factors, the protection is substantial. Practitioners seeking to assess the framework for a specific investment universe should therefore examine the intra-universe correlation structure as the primary determinant of the level of protection that can reasonably be expected under the governance mechanism.

5. System Implementation

A central Orchestrator (`main.py`) coordinates the pipeline by invoking seven agents in a deterministic sequence and in strict topological order. At each step, it forwards the output contract produced by one agent to the next agent in the chain and preserves the governance record as immutable once Agent 6 has committed its outputs. Importantly, the

Orchestrator contains no domain logic of its own: it does not generate instability signals or compute portfolio weights. This separation of concerns allows the governance logic implemented by Agents 1–6 to be observed, falsified, and audited independently of the environment in which the pipeline is deployed.

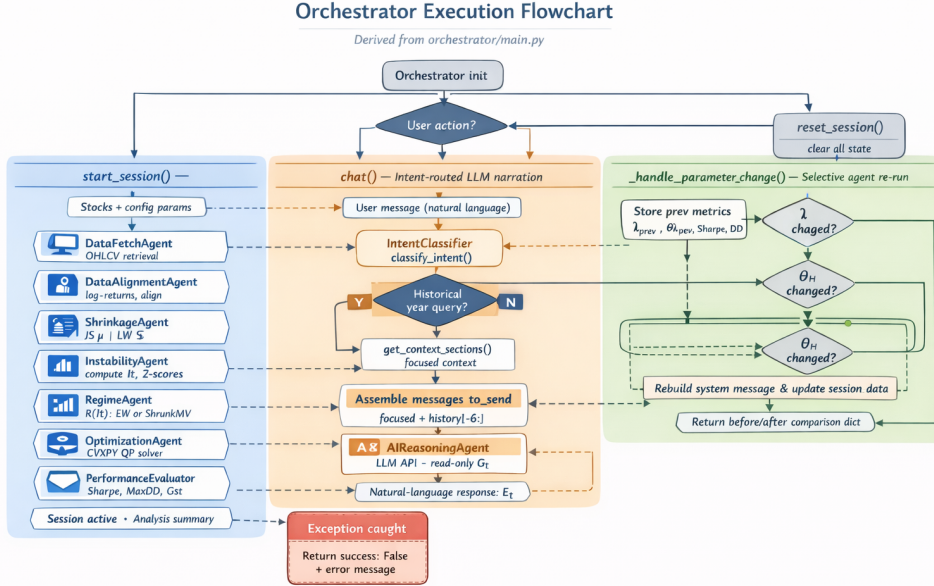


Figure 9: Orchestrator execution flow within the seven-agent pipeline. *Blue path*: Agents A1 through A7 call `start_session()` sequentially, with `session_data` progressively extended by each output. *Orange path*: `chat()` queries are routed through intent classification; requests concerning historical governance records and requests involving parameter changes follow different execution branches before rejoining at the read-only AIReasoningAgent (A8). *Teal path*: `_handle_parameter_change()` re-executes only the agents downstream of the modified parameter: a change to λ re-runs A6 through A7, whereas a change to θ_H re-runs A5 through A7. Red arrows indicate shared exception handling across all execution paths.

5.1. Multi-Agent Pipeline Architecture

The governance framework is instantiated as a seven-agent deterministic Python pipeline in which each agent encapsulates a precisely defined computational task and communicates exclusively via a single immutable data contract. No agent references the internal state of any non-adjacent agent, and no agent is permitted to modify the committed output of any predecessor.

The architecture follows the blackboard pattern as formalised by Wooldridge (2009), adapted for a strictly forward-only, deterministic execution domain. This adaptation eliminates the race conditions and non-reproducible execution sequences that are inherent in conventional reactive blackboard systems. Within each governance cycle, data flows sequentially through Agents 1 to 6; the pipeline state at each checkpoint is a serialisable Python dictionary that any third party can reproduce from the raw price history and the four governing parameters $\{W, \theta_H, \lambda, TC\}$ alone. Full reproducibility from these inputs is the primary engineering objective of the design, and is the property that allows the pipeline to satisfy the audit trail requirements of MiFID II (European Parliament, 2014), GDPR Article 22 (European Parliament, 2016), and Basel III (Basel Committee on Banking Supervision, 2011).

Table 10: Agent roles and data contracts in the seven-agent supervisory governance pipeline, coordinated by the central Orchestrator. The Orchestrator enforces topological execution order and state immutability; it holds no domain logic. Agent 7 (LLM Interface) has strictly read-only access to the committed governance record $\mathcal{G}_t$.

Agent	Primary Function	Input	Output
0. Orchestrator	Sequence agents; pass state contracts; enforce record immutability	$\{W, \theta_H, \lambda, TC\}$; price history P	Coordinated execution; error log
1. DataAlignment	Price retrieval; log-return computation; forward-fill alignment	Raw OHLCV series	Aligned return matrix R
2. Shrinkage	James-Stein return shrinkage; Ledoit-Wolf covariance estimation	R ; training baseline $\mathcal{B}$	$\hat{\mu}_{JS}, \hat{\Sigma}_{LW}$
3. Instability	Composite Instability Index I_t computation and Z-scoring	R ; baseline $\mathcal{B}$	Scalar I_t ; Z-scores $z_\sigma, z_\rho, z_\delta$
4. Regime	Governance rule evaluation via $\mathcal{R}(I_t)$	Scalar I_t ; threshold θ_H	Binary: SHRUNKMV or EQUALWEIGHT
5. Optimisation	Weight computation consistent with active regime	Binary instruction; $\hat{\mu}_{JS}, \hat{\Sigma}_{LW}, \lambda$	$w_t \in \Delta^{p-1}$
6. Performance	Out-of-sample metrics and Governance Stability	w_t, w_{t-1} , realised r_t	Sharpe, MaxDD, Calmar, GS_t
7. LLM Interface	Post-hoc plain-language narration	$\mathcal{G}_t$ (read-only)	Natural-language $\mathcal{E}_t$

Figure 10 illustrates the pipeline as a directed acyclic graph distinguishing the deterministic governance core (Agents 1–6) from the architecturally isolated LLM Interface (Agent 7). The full governance cycle completes in approximately 2.3 seconds on standard consumer hardware.

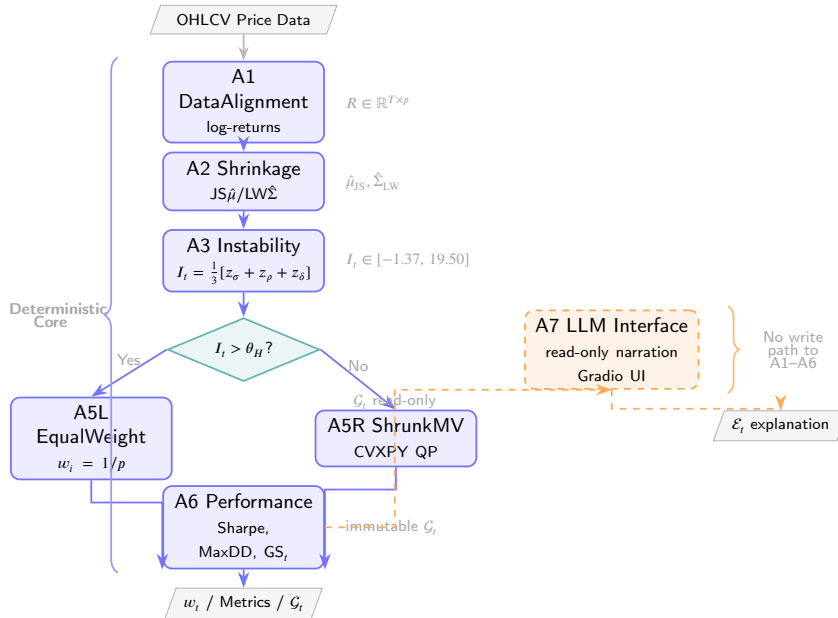


Figure 10: Seven-agent supervisory governance pipeline represented as a directed acyclic graph. Agents A1 through A6 (solid blue) form the deterministic core of the pipeline. Agent A4 (teal diamond) implements the Regime Operator: if $I_t > \theta_H$, execution is routed to the EqualWeight branch (A5L); otherwise, the ShrunkMV branch (A5R) is activated. Both branches converge at Agent A6, where the immutable governance record $\mathcal{G}_t$ is committed. Agent A7 (dashed orange) holds read-only access to $\mathcal{G}_t$ and has no upstream write path into the deterministic core; its isolation from the computation is enforced by the pipeline architecture rather than by prompt-level instruction.

5.2. Conversational Governance Interface

Agent A7 serves as an on-demand natural-language explanation interface for governance decisions. The governance record, comprising the numerical values of I_t , regime flags, and portfolio weight vectors, is formally complete but operationally difficult to interpret for compliance officers and clients without quantitative training. The conversational layer addresses this by translating the committed pipeline state into plain language while leaving all underlying numerical values unchanged.

Each invocation of the language model is grounded in a read-only context $\mathcal{G}_t = \{I_t, \theta_H, \text{mode}, w_t, \text{metrics}\}$. The separation from the decision-making pipeline is exact and verifiable: the API call is issued only after w_t has been written to the governance log, and the model output is routed exclusively to the user interface, with no write path to any upstream component. Because this isolation is enforced by the execution order of the pipeline rather than by prompt-level instruction, it is not susceptible to prompt injection or adversarial override.

A lightweight keyword classifier assigns each incoming query to one of four intent categories: *regime*, *performance*, *counterfactual*, and *allocation*. The classifier then restricts the grounded context $\mathcal{G}_t$ to the fields relevant to the recognised intent, thereby limiting context length and reducing the risk of hallucination. Mean response latency across the four intent categories is 1.4 seconds.

5.3. Representative Governance Cycle: April 2020

The April 2020 evaluation window exhibits the highest level of I_t observed across the twelve-year sample and provides the clearest illustration of how the detection, decision, and explanation layers operate jointly under severe market stress.

Agent A3 — Instability Computation. The three normalised component signals for this window are:

$$z_\sigma = 3.21, \quad z_\rho = 4.87, \quad z_\delta = 8.44 \quad (10)$$

yielding a Composite Instability Index of:

$$I_t = \frac{3.21 + 4.87 + 8.44}{3} = 5.99 \quad (11)$$

This value exceeds the governance threshold $\theta_H = 1.0$ by 4.99 standard deviations. The covariance drift component z_δ is the dominant contributor, reflecting an unprecedented structural rotation in the covariance matrix during the market crash of March 2020.

Agent A4 — Regime Decision. Since $I_t = 5.99 > \theta_H = 1.0$, the Regime Operator outputs *EqualWeight*.

Agent A5-L — Weight Computation. The equal-weight allocation assigns:

$$w_i = \frac{1}{19} = 5.26\% \quad \text{for all } i = 1, \dots, 19 \quad (12)$$

Agent A6 — Governance Commit. The governance record $\mathcal{G}_t$ is committed to the immutable log.

Agent A7 — Audit Narrative. The *AIRReasoningAgent* generates the following governance narrative on demand:

The portfolio was managed under the *EqualWeight* regime for the period ending 31 March 2020. The Composite Instability Index I_t stood 5.99 standard deviations above its 2010–2019 training baseline, with the result driven primarily by covariance matrix drift ($z_\delta = 8.44\sigma$) and higher mean pairwise correlation ($z_\rho = 4.87\sigma$). Under Definition 2, any value of I_t above $\theta_H = 1.0$ unconditionally activates the equal-weight capital protection rule. This is a deterministic process: given the price history up to 31 March 2020 and $\theta_H = 1.0$, *EqualWeight* is the unique and fully reproducible output of the Regime Operator. The dominant driver of the decision is covariance drift, meaning that the co-movement structure of the portfolio is changing at an abnormally rapid rate. That shift weakens the reliability of any covariance-based return estimate and therefore produces a regime transition from *ShrunkMV* to *EqualWeight*.

The narrative above satisfies all four properties of a compliant audit record and may be included in a supervisory report without further modification. It can be reproduced exactly at any future date by re-running the pipeline with the same input price history and parameter set $\{W, \theta_H, \lambda, TC\}$.

The narrative above satisfies all four properties of a compliant audit record and is suitable for inclusion in a supervisory report without further modification. It can be regenerated exactly at any future date by re-running the pipeline with the same input price history and parameter set $\{W, \theta_H, \lambda, TC\}$.

6. Conclusion

6.1. Summary of Findings

This paper introduces a supervisory governance framework that combines statistically grounded drawdown protection, fully auditable allocation decisions, and on-demand plain-language explanations, and evaluates the framework on real-world data spanning a twelve-year period containing multiple stress episodes.

The Composite Instability Index I_t is constructed from cross-sectional volatility, mean pairwise correlation, and the Frobenius norm drift of the covariance matrix. To the best of the authors' knowledge, the drift term is the first use of this quantity as a real-time governance signal for portfolio management. Among the three components, it delivers the strongest separation between market states, with a mean-level ratio of 6.02, and it was the principal driver of the April 2020 peak ($z_\delta = 8.44\sigma$). The lag-1 autocorrelation of 0.84 indicates that I_t captures persistent market regimes rather than transitory noise.

The Regime Operator $\mathcal{R}(I_t)$ takes the index as input and produces one of two portfolio-construction instructions from a single scalar comparison with θ_H . Because the rule is deterministic and binary, its output can be reproduced uniquely from the input price history and the threshold parameter alone, thereby satisfying the auditability requirements of MiFID II Article 25 (European Parliament, 2014), GDPR Article 22 (European Parliament, 2016), and Basel III (Basel Committee on Banking Supervision, 2011) without discretionary intervention.

The Governance Stability metric measures decision stability independently of return performance. The framework records a mean Governance Stability of $\bar{GS} = 0.377$ compared with 0.801 under Static MV, corresponding to a 53% improvement attributable to the stabilising effect of the governance signal on rebalancing frequency.

Across 51 non-overlapping quarterly windows from January 2012 to October 2024, the framework reduces mean maximum drawdown by 38.1% relative to Static MV ($t = 5.67, p < 0.001$). The result remains statistically significant after excluding the COVID-19 period ($n = 47, p < 0.001$), remains stable over $\lambda \in [1.0, 10.0]$ and $\theta_H \in [0.50, 1.50]$ within a band of 2.0 percentage points, and is robust across all five validation universes. The improvement ranges from +12.4% in Universe U2 (European Country ETFs) to +70.0% in Universe U5 (Multi-Asset), suggesting that the governance benefit increases with the degree of factor independence available within the relevant instrument universe. Replication code and reproducibility notebook:

https://github.com/jithendra259/fintech_chatbotk

6.2. Limitations

COVID-19 Underperformance. The framework records a maximum drawdown deterioration of 15.2% in the March 2020 window. In that episode, the Static MV portfolio was, by chance, positioned in low-beta defensive holdings that cushioned the initial shock, whereas the equal-weight fallback spread exposure more evenly across financials and energy — sectors that were hit especially hard in the first phase of the pandemic. The binary fallback is sensitive to the level of I_t , but it cannot distinguish between sector exposures that are protective for a particular shock and those that intensify it. Adjusting θ_H or λ would therefore not resolve this structural limitation.

Equity-Centric Calibration. The rolling realised covariance estimator does not account for fixed income duration non-stationarity, commodity seasonality in co-movement with other asset classes, or derivatives-driven effective correlations in bonds and commodities. The results for Universe U5 (Multi-Asset) are promising but are not sufficiently robust to support rigorous generalisation of the model to cross-asset portfolios without further adaptation of the estimation procedure.

Transaction Cost Approximation. The flat charge of 10 basis points per rebalance understates actual implementation costs during periods of stress, when bid–ask spreads widen and the equal-weight instruction requires the liquidation of concentrated positions. The reported results should therefore be interpreted as gross returns rather than net returns achievable in a live portfolio.

Fixed Threshold. The threshold $\theta_H = 1.0$ is calibrated once against the 2010–2019 training distribution. Secular shifts in the level of market volatility over time could cause a static threshold to generate incorrect governance signals and potentially amplify underperformance in future regimes. Dynamic recalibration of θ_H would address this concern but would introduce an additional free parameter that would itself require out-of-sample stability testing.

6.3. Future Research

Regime Operator with Smooth Transition. Replacing the binary switching rule with a smooth transition blending function applied to I_t would preserve full audit reproducibility while attenuating the abrupt weight changes that contributed to the COVID-19-related underperformance identified in Section ???. A graduated operator, in which the degree of defensiveness scales continuously with I_t , would allow the framework to respond proportionally to intermediate levels of instability rather than applying a discrete all-or-nothing regime change.

Component Weighting. The current specification assigns equal weight to the three component signals of I_t . Adopting signal combination methodologies in the spirit of Billio et al. (2012) and Acharya et al. (2017) would allow for the derivation of optimal component weights that maximise the predictive content of I_t for drawdown beyond the 6.02-fold mean-level separation observed under equal weighting.

Cross-Asset Covariance Drift. Extending the asset universe in the covariance estimation step to include fixed income and commodity instruments would allow the drift component δ_t to capture changes in equity–bond correlation regimes without requiring any modification to the formal definition of the Regime Operator. This would directly address the equity-centric calibration limitation identified in Section ??.

Retrieval-Augmented Reporting. Incorporating a retrieval mechanism into Agent A7 to identify historically analogous governance cycles would enable decisions to be contextualised against documented precedents, extending the plain language explanation interface through grounded generation mechanisms of the kind proposed by Wu et al. (2023) and Li et al. (2023).

Cross-Strategy Benchmarking. Computing the ℓ_1 weight change for risk parity (Roncalli, 2013), smart beta, and liability-driven investing strategies would yield empirical reference values against which the Governance Stability metric could be benchmarked, establishing whether the improvements documented here are specific to the mean-variance family of strategies or generalise more broadly.

CRedit Authorship Contribution Statement

K J Subramanyam: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review and editing, Visualization, Investigation.

Declaration of Competing Interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data Availability

Replication code, data retrieval scripts, and a reproducibility notebook regenerating all tables and figures are publicly available at https://github.com/jithendra259/fintech_chatbotk. Daily price data are retrieved from Yahoo Finance via the open-source `yfinance` Python library and are freely accessible for research purposes.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used an AI language model to assist with manuscript structuring, $\text{\LaTeX}$ formatting, and language editing. After using this tool, the author reviewed and edited all content as needed and takes full responsibility for the content of the published article.

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